

Computer Networks LAB-01

Experiment Title

Study of LAN Topology, Network Devices, and Address Configuration

Problem Statement

① Draw the Network Topology:

- Identify and draw the LAN topology of your organization (e.g., Star, Bus, Mesh).
- Clearly label all components such as Router, Switch, Access Point, and end devices (PCs, laptops, etc).

Soln. Network Topology

The college campus network follows a Hybrid Topology which is a combination of Star Topology and Bus Topology.

In this network, the Main Router and Core Switch are connected in a star structure with floor switches. Each floor contains two switches for proper network distribution. After the switches, multiple Access Points and LAN devices are connected through a linear bus connection.

The server room is located on the second floor and contains the Core Switch and Servers. The network provides both wired (Ethernet) and wireless (Wi-Fi) communication across classrooms, labs, and offices.

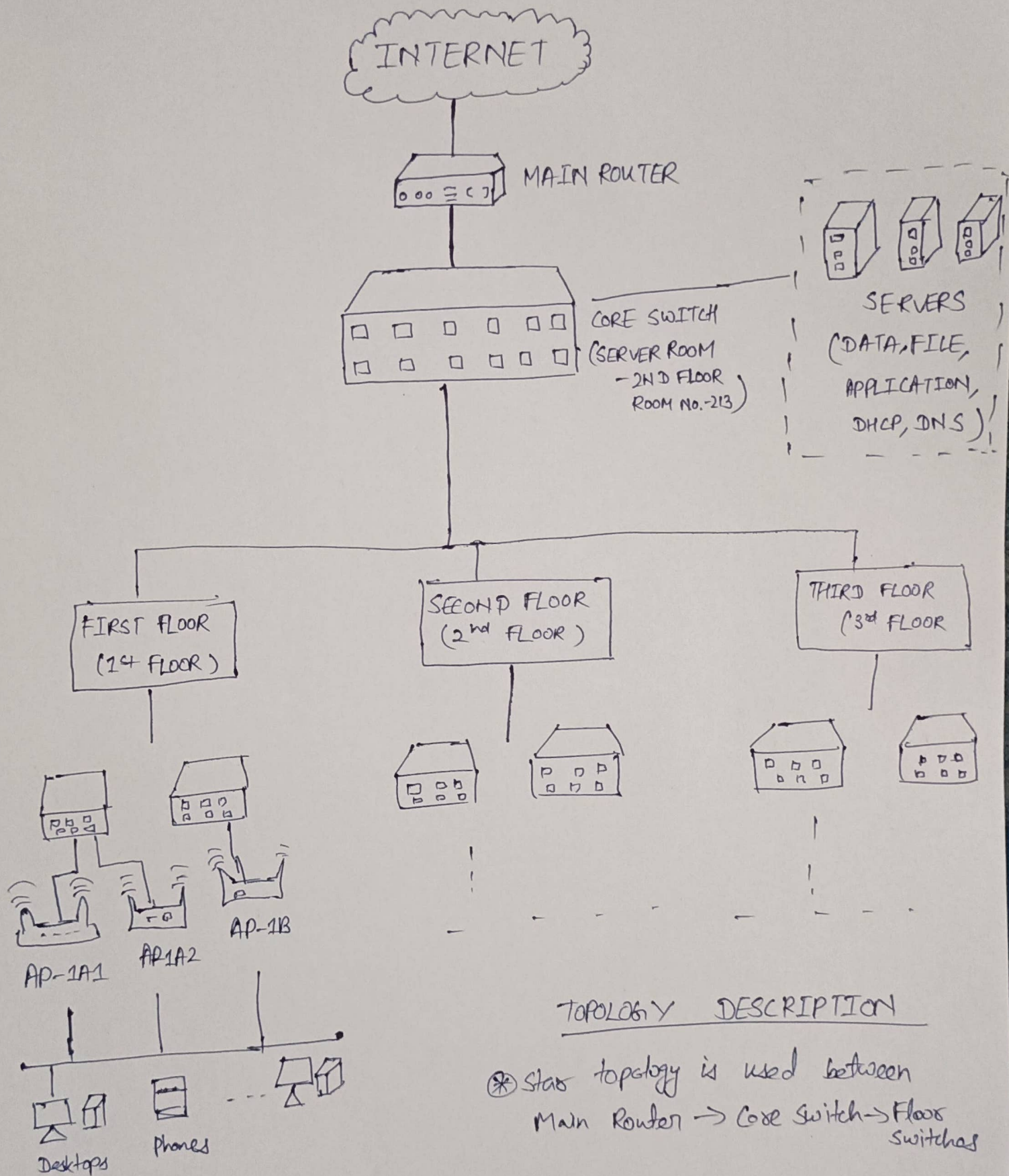
Topology Used:

- Star Topology (between Router, Core Switch, and Floor Switches)
- Bus Topology (between Switches, Access Points, and LAN devices)

Components Used:

- Main Router
- Core Switch
- Floor Switches
- Access Points
- Servers
- LAN Devices
- Ethernet Cables

COLLEGE CAMPUS NETWORK TOPOLOGY (STAR + BUS)



TOPOLOGY DESCRIPTION

- * Star topology is used between Main Router → Core Switch → Floor Switches
- * Each floor has two switches for better distribution and load balancing.
- * This is a Hybrid Topology (Star + Bus)

② Identify Network Devices :

- List all networking devices used in the LAN.
- Mention the function of each device

Solⁿ. The following networking devices are used in the college LAN network :

Network Device

Function

Main Router → Connects the college network to the Internet and routes data between different networks.

Core Switch → Acts as the central backbone of the network and connects all floor switches with the server room.

Floor switches → Distribute network connection to different rooms, labs, and access points on each floor.

Access Points (APs) → Provide wireless Wi-Fi connectivity to laptops, mobiles, and wireless devices.

Servers → Store data, manage applications, provide DHCP, DNS, file sharing, and other network services.

Desktop computers / LAN systems → Used by students, faculty, and labs for network communication and Internet access.

<u>Network Device</u>	<u>Function</u>
Ethernet Cables (LAN Cables)	→ Provide wired communication between switches, routers, servers and computers.

Patch Panel / Patch Rack → Organizes and manages network cable connections in the server room.

③ Analyze Connection Types:

- Specify whether connections are:
 - Wired (Ethernet)
 - Wireless (Wi-Fi)
- Mention where each type is used.

Solⁿ The college campus network uses both wired (Ethernet) and wireless (Wi-Fi) connections for communication.

① Wired Connection (Ethernet)

Wired connections use Ethernet (LAN) cables for data communication.

Where Wired Connection is Used

- Computer Labs
- Classroom LAN systems
- Server Room
- Core Switch to Floor Switch connection
- Switch to Desktop PCs
- Office systems

Features of Wired Connection

- High speed communication
- Stable network connection
- Low latency
- More secure than wireless
- Less affected by Interference

Devices connected through Ethernet

- Desktop Computers
- Servers
- Core Switch
- Floor Switches
- Routers

② Wireless Connection (Wi-Fi)

Wireless connections uses radio waves through Access Points (APs).

Where wireless connection is used

- Faculty rooms
- Student areas
- Mobile devices
- Laptops
- Smart devices
- Areas where cable connection is difficult.

Features of Wireless Connection

- Easy mobility
- No physical cable required
- Easy installation
- Flexible access from different locations

Devices Connected through Wi-Fi

- Laptops
- Mobile Phones
- Tablets
- Wireless IoT Devices

Network Structure Used in College

- The Core Switch and Floor Switches are connected using Ethernet cables.
- Multiple Access Points are connected after the floor switches.
- LAN rooms and lab systems mainly use wired Ethernet connection.
- Wireless users connect through Wi-Fi access points installed on each floor.

Conclusion

The college network uses a hybrid communication system where Ethernet provides stable high-speed communication for labs and servers, while Wi-Fi provides wireless access for mobile users across the campus.

④ Measure Network Speed:

- Find the link speed of:
 - wired network (if available)
 - wireless network
- Mention the tools/commands used (e.g. Task Manager, netsh; netsh wlan show interfaces -or, etc).

Solⁿ.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

C:\Users\harsh> netsh wlan show interfaces

There is 1 interface on the system:

Name               : Wi-Fi
Description        : Intel(R) Wi-Fi 6E AX211 160MHz
GUID               : 3d98d74b-ae17-456b-b7d3-859f3809aa32
Physical address   : a0:59:50:b6:a4:e0
Interface type     : Primary
State              : connected
SSID               : realme 8s 5G
AP BSSID           : ca:22:a0:2a:4b:10
Band               : 2.4 GHz
Channel            : 11
Connected Akm-cipher : [ akm = 00-0f-ac:08, cipher = 00-0f-ac:04 ]
Network type       : Infrastructure
Radio type         : 802.11n
Authentication     : WPA3-Personal
Cipher             : CCMP
Connection mode    : Profile
Receive rate (Mbps) : 72.2
Transmit rate (Mbps) : 72.2
Signal             : 96%
Rssi               : -35
Profile            : realme 8s 5G
QoS MSCS Configured : 0
QoS Map Configured  : 0
QoS Map Allowed by Policy : 0

C:\Users\harsh>
```

↳ using netsh wlan show interfaces

Task Manager

Performance

CPU
2% 1.48 GHz

Memory
11.8/15.6 GB (76%)

Disk 0 (C:)
SSD (NVMe)
1%

Wi-Fi
Wi-Fi
S: 144 R: 160 Kbps

GPU 0
Intel(R) Iris(R) Xe Graphi
1%

Wi-Fi

Throughput

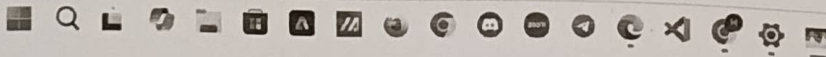
Intel(R) Wi-Fi 6E AX211 160MHz

26 Mbps

17 Mbps

60 seconds

Send	Adapter name:	Wi-Fi
144 Kbps	SSID:	realme 8s 5G
	Connection type:	802.11n
Receive	IPv4 address:	10.234.192.131
160 Kbps	IPv6 address:	2409:40e4:204f:6948:bbae:cf9d:724b:7451
	Signal strength:	



ENG IN 12% 15:4 27-05-2024

↳ In Task Manager → Performance

⑤ Find System Network Details:

- Determine and record the following of your computer:
 - MAC Address
 - IP Address (IPv4)
 - Subnet Mask
- Mention the commands(s) used (e.g. ipconfig, getmac)

Solⁿ.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

C:\Users\harsh> ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40e4:204f:6948:bbae:cf9d:724b:7451
    Temporary IPv6 Address. . . . . : 2409:40e4:204f:6948:b821:8c95:c3d7:d503
    Link-local IPv6 Address . . . . . : fe80::fbc:d858:3ef:1e99%8
    IPv4 Address. . . . . : 10.234.192.131
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::c822:a0ff:fe2a:4b10%8
                                10.234.192.30

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
C:\Users\harsh>
```

↳ using ~~ip~~ ipconfig

IPv4 Address → 10.234.192.131

Subnet Mask → 255.255.255.0

⑥ Explain Observations :

• Briefly explain :

- Difference between wired and wireless speed
- Role of subnet mask in networking

Solⁿ Difference between wired and wireless speed

Wired Network (Ethernet)

- ⊗ Provides faster communication
- ⊗ Connection is stable
- ⊗ Uses ethernet cable
- ⊗ Less interference
- ⊗ Low latency
- ⊗ More Secure

Wireless Network (Wi-Fi)

- ⊗ Speed is comparatively slower
- ⊗ Speed may fluctuate
- ⊗ Uses radio waves
- ⊗ Affected by walls, distance and interference
- ⊗ Higher latency
- ⊗ Less secure compared to wired

Explanation

In the college network, wired Ethernet connection is mainly used in labs, server rooms, and office systems because it provides stable and high-speed communication.

Wireless Wi-Fi connection is used for laptops and mobile devices through Access Points. Wi-Fi provides mobility and easy access, but the speed may decrease due to interference, signal range, and multiple connected users.

Role of Subnet Mask in Network

A subnet mask is used to divide an IP address into:

- Network portion
- Host portion

It helps devices identify whether another device belongs to the same network or a different network.

IP Address : 10.234.192.131

Subnet Mask : 255.255.255.0

Here :

- 10.234.192 represents the network part
- 131 represents the host/device part

Importance of Subnet Mask

- Helps in network identification
- Reduces unnecessary network traffic
- Improves routing efficiency
- Allows proper communication b/w devices